ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)
JEE(Advanced)
FULL SYLLABUS
21-03-2021

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)

PAPER-1

PART-1 : PHYSICS SOLUTION

SECTION-I

1. Ans. (A,B,D)

Sol. Activity of A = λN_A

Activity of B = $2\lambda N_B$

$\lambda N_A = 2\lambda N_B$

$N_A = 2N_B$

2. Ans. (A,B,C,D)

Sol. $w = \frac{\lambda D}{d}$

$n = \left(\frac{xd}{\lambda D}\right)$

$\frac{dn}{dt} = \left(\frac{xv}{\lambda D}\right)$

3. Ans. (A,C,D)

Sol. net force on magnet is zero in uniform magnetic field.

4. Ans. (A,B,C)

Sol. $a = -bx = -w^2x$ $\qquad w = \sqrt{b}$

$V_{max} = aw$ $\qquad A = \frac{u}{\sqrt{b}}$

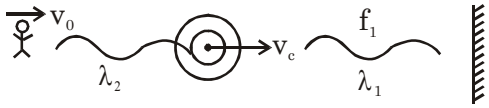
5. Ans. (A,C,D)

Sol. $H_{max} = \frac{2T\cos\theta}{r\rho g} = 1\text{cm}$

$H \propto \cos\theta_C$

6. Ans. (A,B,C,D)

Sol.



$\lambda_1 = (v - v_c)T = \left(\frac{v - v_c}{f}\right)$

$\lambda_2 = (v + v_c)T = \left(\frac{v + v_c}{f}\right)$

$f_1 = \frac{v}{x_1} = \left(\frac{v}{v + v_c}\right)f$

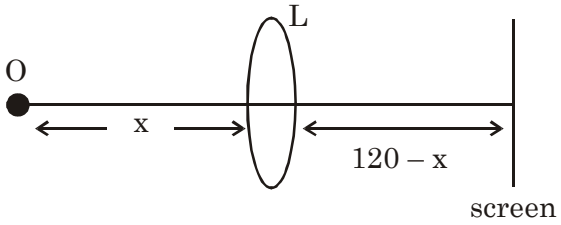
$\Delta f_1 = f_{echo} - f_{street}$

$= \left(\frac{v}{v - v_c}\right)f.\left(\frac{v + v_1}{v}\right) - f\left(\frac{v + v_0}{v + v_c}\right)$

$= \frac{2v_c(v + v_0)f}{(v^2 - v_c^2)}$

7. Ans. (A)

Sol.



$m_1 : m_2 = 1 : 9$

$m_1 . m_2 = 1$

$\therefore \quad m_1 = \frac{1}{3} \qquad m_2 = 3$

$m = \frac{v}{u}$

$\frac{1}{3} = \frac{120 - x_1}{x_1}$

$4x_1 = 360$

$x_1 = 90$

$x_2 = 120 - 90 = 30$

$D = 120$

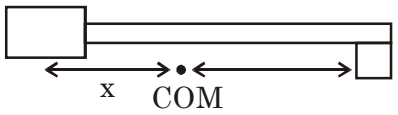
$d = x_1 - x_2 = 60$

$f = \frac{D^2 - d^2}{4D} = 22.5 \text{ cm}$

8. Ans. (A)

9. Ans. (A)

Sol.



$x = \frac{M.L.}{3M + M} = \frac{L}{4}$

from conservation of angular momentum about COM

$$Mv_0\left(\frac{3L}{4}\right) = Iw$$

$$I = 3M\left(\frac{L}{4}\right)^2 + M\left(\frac{3L}{4}\right)^2 = \frac{3ML^2}{16} + \frac{9ML^2}{16}$$

$$= \frac{3}{4}ML^2$$

$$\frac{3Mv_0L}{4} = \frac{3}{4}ML^2.w$$

$$\boxed{w = \frac{v_0}{L}}$$

10. Ans. (B)

Sol. From cons of linear momentum

$Mv_0 = 4M.v_c$

$$v_c = \frac{v_0}{4}$$

Velocity of $3M = v_c - \frac{L}{4}w$

$$= \frac{v_0}{4} - \frac{L}{4} \times \frac{v_0}{L} = 0$$

SECTION-II

1. Ans. 4.80

Sol. $$\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2}$$

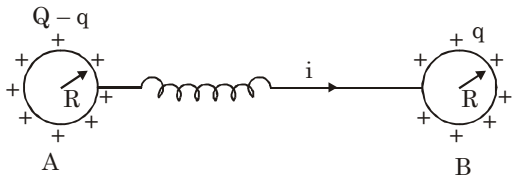
$$t = \frac{t_1t_2}{t_1 + t_2} = \frac{24 \times 6}{24 + 6} = 4.80 \text{ Hr.}$$

2. Ans. 2.82 to 2.83

Sol. Let q = charge on B at time 't'

$$V_A - V_B = L\frac{di}{dt}$$

$$\Rightarrow K\frac{Q-q}{R} - K\frac{q}{R} = L\frac{d^2q}{dt^2}$$



[Spheres are at large distance. Hence charge on one does not affect the potential of other]

$$\Rightarrow \frac{d^2q}{dt^2} = -\frac{2K}{LR}\left[q - \frac{Q}{2}\right] \quad \text{..... (i)}$$

Let $q - \frac{Q}{2} = x$ which means $\frac{d^2q}{dt^2} = \frac{d^2x}{dt^2}$

In terms of x, the differential equation (i) becomes

$$\frac{d^2x}{dt^2} = \frac{2K}{LR}x$$

Solution to this equation is

$$x = x_0\sin(\omega t + \delta) \quad \left[\omega = \sqrt{\frac{2K}{LR}}\right]$$

$$\Rightarrow q - \frac{Q}{2} = x_0\sin(\omega t + \delta)$$

$$\Rightarrow q = \frac{Q}{2} + x_0\sin(\omega t + \delta)$$

Current

$$i = \frac{dq}{dt} = x_0\omega\cos(\omega t + \delta)$$

Just after closing the switch (at $t = 0^+$) the current is zero

$$\therefore \delta = \frac{\pi}{2}$$

$$\therefore q = \frac{Q}{2} + x_0\cos\omega t$$

Also, at t = 0, q = 0

$\therefore x_0 = -Q/2$

$$\therefore q = \frac{Q}{2}\left[1 - \cos\omega t\right]$$

$q = \frac{Q}{2}$ means $\cos\omega t = 0$

$$\Rightarrow \omega t = \frac{\pi}{2}$$

$$\Rightarrow \sqrt{\frac{2K}{LR}}t = \frac{\pi}{2}$$

$$\Rightarrow t = \frac{\pi}{2}\sqrt{\frac{LR}{2K}} = \frac{\pi}{2\sqrt{2}} = \sqrt{\frac{LR}{K}}$$

$\therefore a = 2\sqrt{2}$

$= 2 \times 1.414$

$= 2.828$

$\simeq 2.83$

3. Ans. 60.00

Sol. Initially $V_R = V_{C_1} = 2.5 \times 4 = 10V$

$Q_1 = CV = 3 \times 10 = 30\ \mu C$

$U_1 = \frac{1}{2} \times 3 \times 10^2 = 150\ \mu J$

Finally $V_{C_1} = V_{C_2} = \frac{Q_1}{C_1 + C_2} = \frac{30}{5} = 6\ V$

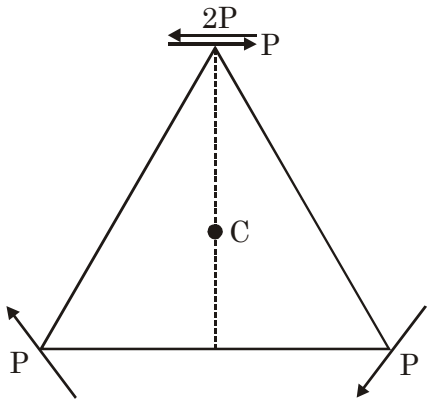
$U_2 = \frac{1}{2} \times (C_1 + C_2) \times V^2$

$= \frac{1}{2} \times 5 \times 36 = 90\ \mu J$

$\therefore$ Heat liberated = 150 – 90 = 60 μJ

4. Ans. 10.38 to 10.40

Sol.



From superposition

$E = \left(\frac{2KP}{r^3}\right)$ opposite to $2\vec{P}$

$= \frac{2KP}{\left(\frac{L}{\sqrt{3}}\right)^3}$ $\qquad r = \frac{2}{3} \times \frac{\sqrt{3}L}{2} = \frac{L}{\sqrt{3}}$

$= 6\sqrt{3}\left(\frac{KP}{L^3}\right)$

$\therefore$ a = 6 × 1.732 = 10.392

5. Ans. 1.84

Sol. Length of 1 main scale division
= 1/10 = 0.1 cm
No. of vernier scale divisions = 10
Main scale reading = 1.8 cm
Coinciding vernier scale division = 4th
Least Count
= Length of smallest main scale division/ No. of vernier scale divisions
= 0.1 /10
= 0.01 cm
(vernier calliper has a least count 0.01 cm. So, measurement is accurate only upto three significant figures.)
length = Main scale reading + coinciding vernier scale division × LC
= 1.8 + 4 × 0.01
= 1.8 + 0.04
= 1.84 cm

6. Ans. 0.00

Sol. Heat lost by water = Heat given by ice
(3 + 1) × 1 × (20 – θ)
= 1 × 0.5 × 40 + 1 × 80 + 1 × 1 × θ
80 – 4θ = 20 + 80 + θ
5θ = 20
Not possible ⇒ θ = 0°

7. Ans. 6.00

Sol. After every collision velocity becomes $\frac{1}{\sqrt{2}}$ times.

$\frac{1}{8} = \left(\frac{1}{\sqrt{2}}\right)^n \Rightarrow n = 6$

8. Ans. 6.00

Sol. Magnitude of viscous force, $F = \eta A = \frac{dv}{dr}$

$\Rightarrow$ viscous force per unit area $\sigma = \frac{F}{A} = \eta\frac{dv}{dr}$

$v = v_0\left(1 - \frac{r^2}{R^2}\right)$

$\Rightarrow \frac{dv}{dr} = -\frac{2V_0 r}{R^2}$

$\Rightarrow \sigma = \eta\frac{2v_0 r}{R^2}$ ..... (i)

Volume rate of flow, Q
consider an annular element at r from axis, width dr.

$dA = 2\pi r dr;\ dQ = v.dA = v_0\left(1 - \frac{r^2}{R^2}\right)2\pi r dr$

$Q = \int dQ = 2\pi v_0\left[\frac{r^2}{2} - \frac{r^4}{4R^2}\right]_0^R = \frac{\pi}{2}R^2 v_0$

$\Rightarrow\ v_0 = \frac{2Q}{\pi R^2}$

$\therefore$ (i) $\Rightarrow \sigma = \eta\frac{4Q}{\pi R^4}r$, R = 0.1 m
At r = 0.04 m,

$\sigma = (0.75)4 \times \frac{\pi}{2} \times 10^{-2} \times \frac{0.04}{\pi \times 10^{-4}}$

= 6 Nm^{-2}

PART–2 : CHEMISTRY SOLUTION

SECTION-I

1. Ans. (A,B,C)

Sol. (A) Only first four spectral lines belonging to Balmer series in hydrogen spectrum lie in visible region.

(B) If a light of frequency ν falls on a metal surface having work function $h\nu_0$, photoelectric effect will take place only if $\nu \geq \nu_0$, since ν_0 is the minimum frequency required for photoelectric effect.

(C) The number of photoelectrons ejected from a metal surface in photoelectric effect depends upon the intensity of incident radiations.

2. Ans. (B,C,D)

Sol. B : Magnesite is ore of Mg

C : Anode mud will have Au only

D : Self reduction occurs during roasting.

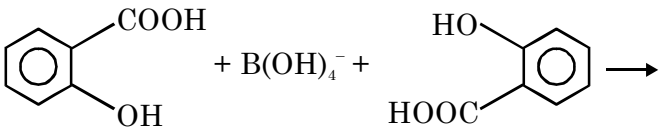
3. Ans. (B,C,D)

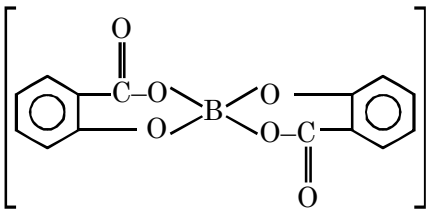
4. Ans. (B,D)

5. Ans. (A,C,D)

Sol. $Na_2B_4O_7 \cdot 10H_2O + aq \rightleftharpoons 4H_2BO_3 + 2NaOH$;

$H_2BO_3 + OH^- \rightleftharpoons B(OH)_4^-$





– Optically resolvable due to asymmetric structure

– Two six membered rings.

6. Ans. (A,B,C)

Sol. Option (D) incorrect because

• Rate of SN^1 depend on stability of carbocation formed.

7. Ans. (B)

Sol.

	A	+ B	→ C	+ D
t=0	1	1	0	0
t=t	0.25	0.25	0.75	0.75

$\frac{dx}{dt} = k(a-x)^{1/2}(b-x)^{1/2}$

Here a = b

$\frac{dx}{dt} = k(a-x)$

$\therefore \quad t = \frac{1}{k}\ln\frac{a}{a-x}$

$\therefore$ We'll treat it as first order, single reactant.

$\therefore \quad t = 2 \times t_{1/2} = 2 \times 0.693 / 2.31 \times 10^{-3} = 600$ s.

8. Ans. (C)

Sol. $\frac{d}{dT}(\ln k) = \frac{E_a}{RT^2}$

$0 + \frac{2500}{T^2} + \frac{3}{T} = \frac{E_a}{RT^2}$

$\therefore \quad E_a = (2500 + 3T)R$

9. Ans. (B)

Sol. $CuSO_4 + KI \rightarrow \underset{\text{white}}{Cu_2I_2}\downarrow + I_2$

$I_2 + Na_2S_2O_3 \rightarrow Na_2S_4O_6 + NaI$

reduced oxidised

10. Ans. (D)

Sol. Bottles 1, 2, 3, 4 have $Pb(NO_3)_2$, HCl, Na_3CO_3 and $CuSO_4$.

When mixing :

Bottle 1 + Bottle 2 → $PbCl_2$

Bottle 1 + Bottle 3 → $PbCO_2$

Bottle 1 + Bottle 4 → $PbSO_4$

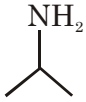
Bottle 2 + Bottle 3 → CO_2

Bottle 2 + Bottle 4 → No visible reaction

Bottle 3 + Bottle 4 → $CuCO_3$

SECTION-II

1. Ans. (23.72)

Sol.: X is 

% Nitrogen = $\frac{14}{59} \times 100 = 23.72\%$

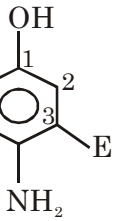
2. Ans. (5.00)

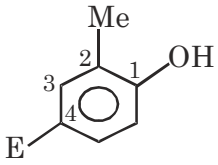
Sol.: 1, 4, 5, 6, 8 can be oxidised.

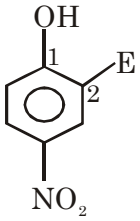
3. Ans. (0.10)

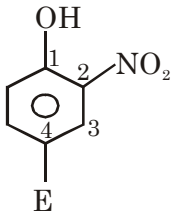
4. Ans. (13.00)

Sol.: 3 + 4 + 2 + 4 = 13

(a) 

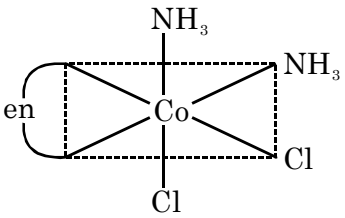
(b) 

(c) 

(d) 

5. Ans. (6.00)

Sol.: (a) Cis-isomer is op. active

(b)  is op. active

(c) optically inactive.

(d) op. inactive

(e) always optically active

(f) square planar, optically inactive

(g) EDTA complexes are optically active

(h) 3 didentate ligands are always optically active

(i) same as (h)

6. Ans. (28.00 to 34.00)

Sol.: $\frac{(In^-)}{(HIn^-)} = \frac{25}{75} = \frac{1}{3}$

$pH = 9.6 + \log \frac{(In^-)}{(HIn^-)}$

$= 9.6 + \log \frac{1}{3}$

$pH = 9.12 = x$

$NaOH + HCl \rightarrow NaCl + H_2O$

1 1

$Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$

1.m.mole	x	0	0
1– x	0	x	–

$pH = pka_2 + \log \frac{(CO_3^{-2})}{(HCO_3^-)}$

$9.12 = 11 + \log \frac{1-x}{x}$

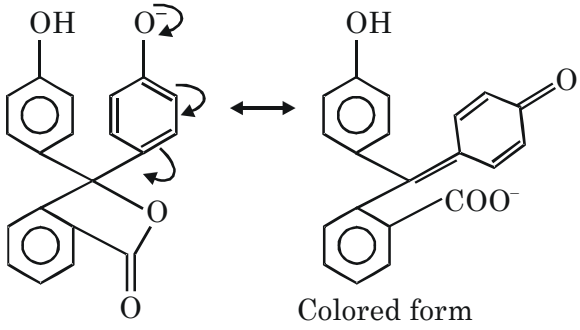
$1.88 = \log \frac{x}{1-x}$

$75.85 = \frac{x}{1-x}$

$x = \frac{75.85}{76.85} = 0.99$

$\therefore\ 0.99 = 0.1 \times V_{HCl}$

$V_{HCl\ Total} = 9.9 + 10 = 19.9\ ml = y$



Colored form

$\therefore\ z = 0$

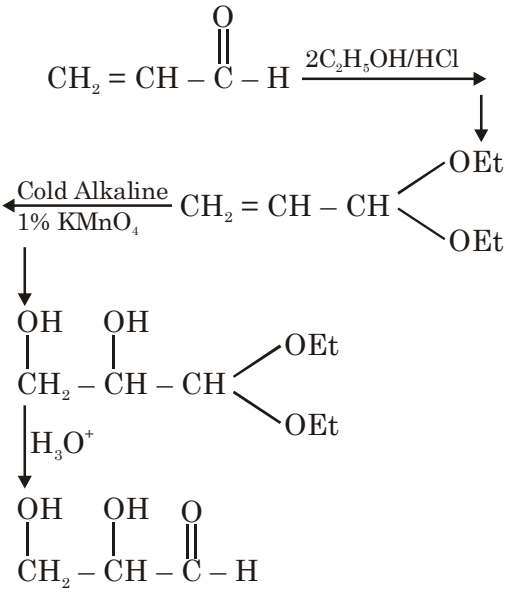
$\therefore\ x + y + z = 9.12 + 19.9 + 0$

$= 29.02$

7. Ans. (6.00)

Sol.: a, b, c, f, h, i

8. Ans. (56.00)

Sol.: 

PART-3 : MATHEMATICS SOLUTION

SECTION-I

1. Ans. (B,C,D)

Sol. $\Delta = \begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$ (for Non trivial solution)

$-(p + q + r)(p^2 + q^2 + r^2 - pq - qr - rp) = 0$

$p + q + r = 0$

Represents line $x = y = z$

$p^2 + q^2 + r^2 - pq - qr - rp = 0$

$\frac{1}{2}(p - q)^2 + (q - r)^2 + (r - p)^2 = 0$

p, q, r must be equal $p = q = r$

$\Rightarrow$ identical planes $x + y + z = 0$

2. Ans. (B,C)

Sol. $-2 < \log_{\frac{1}{\sqrt{3}}}\left(\frac{|z|^2 - |z| + 1}{2 + |z|}\right) < 2$

$\left(\frac{1}{\sqrt{3}}\right)^2 < \frac{|z|^2 - |z| + 1}{2 + |z|} < \left(\frac{1}{\sqrt{3}}\right)^{-2}$

$\Rightarrow \frac{1}{3} < \frac{|z|^2 - |z| + 1}{2 + |z|} < 3$

Let $|z| = t \geq 0$

$\frac{1}{3} < \frac{t^2 - t + 1}{2 + t}$

$2 + t < 3t^2 - 3t + 3$

$\Rightarrow 3t^2 - 4t + 1 > 0$

$\Rightarrow (t - 1)(3t - 1) > 0$

$t < \frac{1}{3}$ or $t > 1$ ...(i)

$\frac{t^2 - t + 1}{2 + t} < 3$

$\Rightarrow t^2 - 4t - 5 < 0$

$\Rightarrow (t + 1)(t - 5) < 0$

$\Rightarrow -1 < t < 5$ ...(ii)

From (i) and (ii)

$t \in \left[0, \frac{1}{3}\right) \cup (1, 5) \Rightarrow 0 \leq |z| < \frac{1}{3}$ or $1 < |z| < 5$

3. Ans. (A,B,C,D)

Sol. Let $B \equiv (2t^2, 4t)$, then $C \equiv (2t^2, -4t)$

then $-2 = 2 + (-t) \Rightarrow t = 1$

$B = (2, 4)$, $C = (2, -4)$

or $ty = x + 2t^2$

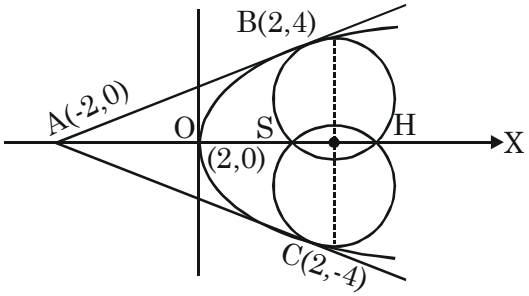
Put $(-2, 0)$

$0 = -2 + 2t^2$

$t = \pm 1$

$\therefore$ $(2t^2, 4t)$

$= (2, 4), (2, -4)$



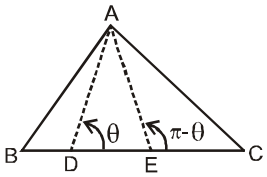
Clearly AB = AC, therefore AS is a median of ΔABC.

It is also the internal angular bisector $\angle BAC$ and it is also $\perp$ from A to opposite side BC. AS is also perpendicular bisector of side BC.

Hence, centroid, incentre, circumcentre and orthocentre lie on AS and hence also on common chord SH of the two circles.

4. Ans. (A,C, D)

Sol.



if we apply m-n Rule in Δ ABE, we get

$(1+1)\cot\theta = 1.\cot B - 1.\cot\theta$

$2\cot\theta = \cot B - \cot\theta$

$3\cot\theta = \cot B$

$\tan\theta = 3\tan B$ .....(1)

Similarly, if we apply m-n Rule in Δ ACD, we get

$(1+1)\cot(\pi - \theta) = 1.\cot\theta - 1.\cot C$.

$\cot C = 3\cot\theta \Rightarrow \tan\theta = 3\tan C$ .....(2)

form (1) and (2) we can say that

$\tan B = \tan C \quad \Rightarrow \quad B=C$

$\because A + B + C = \pi$

$\therefore A = \pi - (B + C)$

$= \pi - 2B \quad \because B = C$

$\therefore \tan A = -\tan 2B$

$$= -\left(\frac{2\tan B}{1-\tan^2 B}\right) = -\frac{\frac{2\tan\theta}{3}}{1-\frac{\tan^2\theta}{9}}$$

$$\Rightarrow \tan A = \frac{6\tan\theta}{\tan^2\theta - 9}$$

5. Ans. (A,B,D)

Sol. Characteristics equation for A :

$\det(A - \lambda I_3) = 0$

$$\begin{vmatrix} 1-\lambda & 2 & 2 \\ 2 & 1-\lambda & 2 \\ 2 & 2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - 9\lambda - 5I = 0$$

Characteristics equation

$\Rightarrow A^3 - 3A^2 - 9A - 5I = 0$

To determine inverse :

$(A^3 - 3A^2 - 9A - 5I)A^{-1} = 0$

$A^2 - 3A - 9I - 5A^{-1} = 0$

$\Rightarrow 5A^{-1} = A^2 - 3A - 9I$

$$A^{-1} = \frac{1}{5}[A^2 - 3A - 9I]$$

$$A^2 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

[For the matrix to be invertible $\det A^2 \neq 0$]

$\det A^2 \neq 0$

$\Rightarrow A^2$ is invertible.

6. Ans. (A,C,D)

Sol. (A) Applying $\int_a^b f(a+b-x)dx$

$$I = \int_a^{\pi-a} (\pi - x)f(\sin(\pi - x))dx = \int_a^{\pi-a} (\pi - x)f(\sin x)dx$$

$$= \pi\int_a^{\pi-a} f(\sin x)dx - \int_a^{\pi-a} xf(\sin x)dx$$

$$I = \pi\int_a^{\pi-a} f(\sin x)dx - I$$

$$\Rightarrow \quad 2I = \pi\int_a^{\pi-a} f(\sin x)dx$$

$$I = \frac{\pi}{2}\int_a^{\pi-a} f(\sin x)dx$$

(B) $\int_{-a}^{a} f^2(x)dx = 2\int_0^a f^2(x)dx$ is true only when f(x) is either even or odd.

(C) $\cos^2 x$ is periodic with fundamental period π

$$\int_0^{n\pi} f(\cos^2 x)dx = n\int_0^{\pi} f(\cos^2 x)dx$$

(D) Put $x + c = t \Rightarrow dx = dt$

$x = 0, \ t = c$

$x = b - c \quad t = b$

$$\int_0^{b-c} f(x+c)dx = \int_c^b f(t)dt = \int_c^b f(x)dx$$

Paragraph for Question 7 and 8

7. Ans.(A)

8. Ans.(C)

Sol. $f(x) = x^3(x-2)^2(x-1)$

$f'(x) = x^3(x-2)^2 + 2x^3(x-2)(x-1) + 3x^2(x-2)^2(x-1)$

$= x^2(x-2)(x^2 - 2x + 2x^2 - 2x + 3x^2 - 9x + 6)$

$= x^2(x-2)(6x^2-13x+6)$

$f'(x) = x^2(x-2)(2x-3)(3x-2)$

$\Rightarrow$ maxima at $x=\frac{3}{2}$ & minima at $x=\frac{2}{3},2$

$\sum x_i^2 = \frac{9}{4}$ (For maxima)

$f(2) = 0$

$f\left(\frac{2}{3}\right)=\frac{8}{27}\times\frac{16}{9}\times\left(-\frac{1}{3}\right)=-\frac{128}{729}$

9. Ans. (C)

10. Ans. (B)

Sol. Given function f(x) can be rewritten as,

$$f(x)=\begin{cases}0, & x<-1\\ 1+x, & -1\le x\le 0\\ 1-x, & 0<x\le 1\\ 0, & x>1\end{cases}$$

$\Rightarrow$
$$f(x-1)=\begin{cases}0, & x-1<-1\\ 1+(x-1), & -1\le x-1\le 0\\ 1-(x-1), & 0<x-1\le 1\\ 0, & x-1>1\end{cases}$$

or
$$f(x-1)=\begin{cases}0, & x<0\\ x, & 0\le x\le 1\\ 2-x, & 1<x\le 2\\ 0, & x>2\end{cases}$$

Also
$$f(x+1)=\begin{cases}0, & x+1<-1\\ 1+(x+1), & -1\le x+1\le 0\\ 1-(x+1), & 0<x+1\le 1\\ 0, & x+1>1\end{cases}$$

or
$$f(x+1)=\begin{cases}0, & x<-2\\ 2+x, & -2\le x<-1\\ -x, & -1<x\le 0\\ 0, & x>0\end{cases}$$

Now, $g(x) = f(x-1) + f(x+1)$

$$=\begin{cases}0, & x<-2\\ 2+x, & -2<x\le -1\\ -x, & -1<x\le 0\\ x, & 0<x\le 1\\ 2-x, & 1<x\le 2\\ 0, & x>2\end{cases}$$

It is easy to check that g(x) is continuous for all $x \in R$ and non-differentiable at $x = -2, -1, 0, 1, 2$

SECTION-II

1. Ans. (27.00)

Sol.: Let O, $A(\vec{a}), B(\vec{b}), C(\vec{c})$ be the vertices of given tetrahedron

$V = \frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}]$

The centroid are $\frac{\vec{a}+\vec{b}}{3}, \frac{\vec{b}+\vec{c}}{3}, \frac{\vec{c}+\vec{a}}{3}, \frac{\vec{a}+\vec{b}+\vec{c}}{3}$

$$V' = \frac{1}{6}\left[\frac{\vec{c}-\vec{a}}{3}\ \frac{\vec{c}-\vec{b}}{3}\ \frac{\vec{c}}{3}\right] = \frac{1}{6\times 27}[\vec{a}\ \vec{b}\ \vec{c}] = \frac{1}{27}V$$

$\therefore$ $k = 27$

2. Ans. (6.82 or 6.83)

Sol.: $\because$ Equation of circle

$(x-2)^2 + (y+2)^2 + \lambda(x+y) = 0$ ...(i)

$\because$ Centre lies on the x-axis

$\therefore$ $\lambda = -4$ put in (i)

$\therefore$ equation of circle is $x^2 + y^2 - 8x + 8 = 0$

(α, β) lies on it

$\Rightarrow$ $\beta^2 = -\alpha^2 + 8\alpha - 8 \ge 0$

$\therefore$ greatest value of 'α' is $4 + 2\sqrt{2}$.

3. Ans. (13.66 or 13.67)

Sol.: $(2x^3 - y)dx + (x + 2yx^2)dy = 0$

$$2x\,dx + 2y\,dy + \frac{x\,dy - y\,dx}{x^2} = 0$$

i.e. $d(x^2 + y^2) + d\left(\frac{y}{x}\right) = 0$

$\therefore$ $d\left(x^2 + y^2 + \frac{y}{x}\right) = 0$

$\therefore$ $p(x, y) = x^2 + y^2 + \frac{y}{x}$

Now $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z+1}{1} = \lambda$

$\therefore x = 2\lambda + 1,\ y = 3\lambda - 1,\ z = \lambda - 1 = 0$

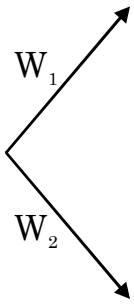
$\Rightarrow \lambda = 1$

$\therefore x = 3, y = 2$

$\therefore c = x^2 + y^2 + \frac{y}{x} = 9 + 4 + \frac{2}{3} = \frac{41}{3}$

4. Ans. (7.00)

Sol.:



$$\frac{1}{2} \xrightarrow{A} \frac{\frac{7!}{2!}+\frac{7!}{2!}-6!}{\frac{8!}{2!\,2!}}$$

$$= \frac{4(7!-6!)}{8!} = \frac{24.6!}{8!} = \frac{3}{7}$$

$$\frac{1}{2} \xrightarrow{A} \frac{7!}{\frac{8!}{2!}} = \frac{2\times 7!}{8!} = \frac{1}{4}$$

$$\text{Required probability} = \frac{\frac{1}{2}\times\frac{3}{7}}{\frac{1}{2}\times\frac{3}{7}+\frac{1}{2}\times\frac{1}{4}}$$

$$= \frac{12}{19} = \frac{p}{q}$$

$\therefore \quad q - p = 7$

5. Ans. (80.00)

Sol.: $\vec{a}_1 = (a, b),\ \vec{a}_2 = \left(a-2, b-\sqrt{5}\right),\ \vec{a}_3 = \left(a+2, b-\sqrt{5}\right)$

Area of Δ, whose vertices are $\vec{a}_1, \vec{a}_2$ and $\vec{a}_3$, is

$$\Delta = \frac{1}{2}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}$$

$$\Rightarrow \Delta^2 = \frac{1}{4}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix} \text{ and}$$

$$\left|\vec{a}_1 - \vec{a}_2\right| = 3,\ \left|\vec{a}_1 - \vec{a}_3\right| = 3,\ \left|\vec{a}_2 - \vec{a}_3\right| = 4$$

$$\Rightarrow \Delta = \sqrt{5\times 2\times 2\times 1} = 2\sqrt{5}$$

$$\left(2\sqrt{5}\right)^2 = \frac{1}{4}\begin{vmatrix} 1+\vec{a}_1\cdot\vec{a}_1 & 1+\vec{a}_1\cdot\vec{a}_2 & 1+\vec{a}_1\cdot\vec{a}_3 \\ 1+\vec{a}_2\cdot\vec{a}_1 & 1+\vec{a}_2\cdot\vec{a}_2 & 1+\vec{a}_2\cdot\vec{a}_3 \\ 1+\vec{a}_3\cdot\vec{a}_1 & 1+\vec{a}_3\cdot\vec{a}_2 & 1+\vec{a}_3\cdot\vec{a}_3 \end{vmatrix}$$

$\therefore$ Required answer = $4 \times 5 \times 4 = 80$

6. Ans. (2.00)

Sol.: $AB - BP = 2I,\ B = \text{adj}(B)$

Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

$$\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$\therefore\ b = c = 0$ and $a = d$

$$\therefore\ B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$$

$aA - aP = 2I$

$aA = aP + 2I$

Let $P = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ where $\begin{matrix} \alpha\delta - \beta\gamma = 3 \\ \alpha + \delta = 4 \end{matrix}$

$$\therefore aA = \begin{bmatrix} a\alpha + 2 & a\beta \\ a\gamma & a\delta + 2 \end{bmatrix}$$

$\Rightarrow a^2|A| = a^2(\alpha\delta - \beta\gamma) + 2a(\alpha + \delta) + 4$

$\Rightarrow a^2|A| = 3a^2 + 8a + 4$

$|A| = 15 \qquad T_r(B) < 0 \Rightarrow a < 0$

$\therefore\ 12a^2 - 8a - 4 = 0$

$a = -\frac{1}{3},\ 1 \Rightarrow a = 1\ (\because\ a < 0)$

$\therefore \quad \text{Tr}(B) = 2a = -2/3$

7. Ans. (16.00)

Sol.: $a_1 = 15,\ a_2 = 15 + d,\ a_3 = 15 + 2d$ .....

$a_{10} = 15 + 9d$

$$\sum_{r=1}^{10} a_r^2 = 15^2 + (15+d)^2 + \ldots + (15+9d)^2$$

$$= 1185$$

$15^2 \times 10 + d^2(1^2 + 2^2 + \ldots + 9^2)$

$+ 30d(1 + 2 + \ldots + 9) = 1185$

$$2250 + d^2 \times \frac{9\times 10\times 19}{6} + 30d \times 45 = 1185$$

$15 \times 19\ d^2 + 30 \times 45d + 1065 = 0$

$19d^2 + 90d + 71 = 0$

$(19d + 71)(d + 1) = 0$

$\therefore \quad d = -1,\ \frac{-71}{19}$ (as a_2 is integer) (rejected)

$\therefore \quad S(n) - S(n-1) \geq 0$

$\Rightarrow T_n \geq 0$

$\therefore \quad n = 16$

8. Ans. 7.00

$S^2 = (1 + 3x + 5x^2 + \ldots + 21x^{10})$

$(1 + 3x + 5x^2 + \ldots + 21x^{10})$

Coeff. of $x^{18} = 21.17 + 19.19 + 17.21 = 1075$

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21-03-2021

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)

PAPER-2

PART-1 : PHYSICS SOLUTION

SECTION-I

1. Ans. (A,D)

Sol. $\pi r^2\sqrt{2gy} = -\pi x^2\frac{dy}{dx}$

$$r^2\sqrt{2gy}\,dt = -\frac{y^{2/a}}{k^{2/a}}dy$$

$$\boxed{k^{2/a}r^2\sqrt{2g}\int_0^t dt = +\int_4^{16} y^{\frac{2}{a}-\frac{1}{2}}dy}$$

(A) $k = 1,\ a = 4$

$\sqrt{5}\times10^{-4}\sqrt{20}\,t = 12$

$t = 12000$ sec

(B) $k = 1,\ a = 2$

$$r^2\sqrt{2g}\,t = t\int_4^{16} y^{\frac{1}{2}}\,dy$$

$$r^2\sqrt{2g}\,t = \frac{2^4\left(16^{3/2}-4^{3/2}\right)}{3}$$

$$\sqrt{5}\times10^{-4}\sqrt{20}\,t = \frac{2}{3}(56)$$

$$t = \frac{2(56)\times10^4}{3\sqrt{5}\sqrt{20}}$$

$$= \frac{2}{3}(56)\times1000$$

(C) $(4)\sqrt{5}\times10^{-4}\sqrt{20}t = \frac{2}{3}(56)$

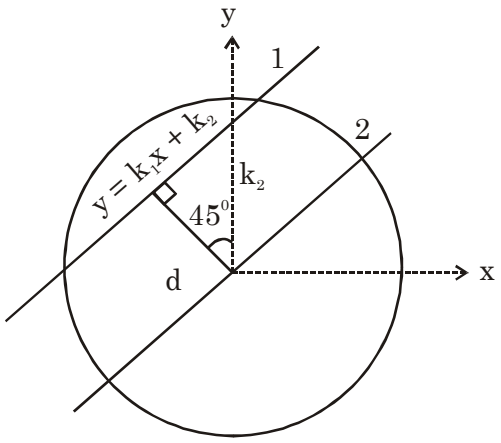
$$t = \frac{2(56)\times1000}{3\times4}$$

(D) $\sqrt{4}\sqrt{5}\times10^{-4}\sqrt{20}\,t = \frac{2}{3}(56)$

$t = 6000$ sec

2. Ans. (B,C)

Sol.



$I_1 = I_2 + md^2$

$$I_1 = \frac{MR^2}{2} + M\left(\frac{K_2}{\sqrt{2}}\right)^2$$

(A) $I_1 = \frac{MR^2}{2} + \frac{MR^2}{4}$

$$= \frac{3MR^2}{4}$$

(B) $I_1 = \frac{MR^2}{2} + \frac{MR^2}{2} = MR^2$

(C) $I_1 = \frac{MR^2}{2} + \frac{MR^2}{2} = MR^2$

(D) $I_1 = \frac{MR^2}{2} + \frac{MR^2}{4} = \frac{3MR^2}{4}$

3. Ans. (A,B,D)

Sol. Power leaving the system = P

$P = \sigma(4\pi C^2)T_C^4$

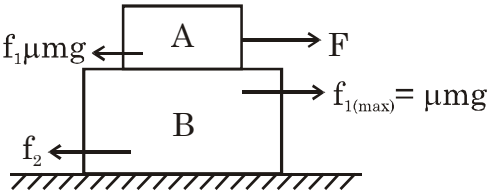
Net power leaving B = P

$P = \sigma(4\pi b^2)T_B^4 - \sigma(4\pi b^2)T_C^4$

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4. Ans. (A,B,C)

Sol.



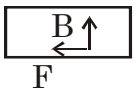
$f_{(max)} = 3\mu mg$

Block 'B' always remains at rest as net force on B is zero.

$$F_{GROUND} = \sqrt{f^2 + \mu}$$

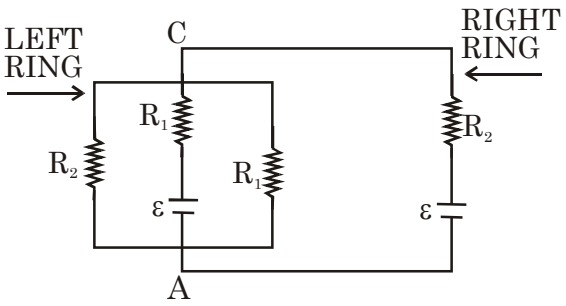
$$= \sqrt{(\mu mg)^2 + (3mg)^2}$$

$$= \boxed{mg\sqrt{9+\mu}}$$



5. Ans. (A,D)

Sol. Simplified diagram may be



So there will be current in each ring and so magnetic force on them will be nonzero.

6. Ans. (A,D)

Sol. $30 = \frac{\varepsilon_1}{20}(4)$

$\varepsilon_1 = 150$

$\varepsilon_2 = \frac{\varepsilon_1}{20} \times 6 = 45V$

7. Ans. (B)

Sol. I $P \propto V$

w^+

$\frac{I}{V} \propto V$

$T \propto V^2$

U = +ve

Q = 4

II. w^+

$\Delta U = 0$

$\theta = +$

III $\Delta U = 0$

w^-

$\theta = -$

IV w^-

$U = -\ V \propto T$

$\theta = -w$

I A,C,D

II A,B,D

III B,E

IV B,E

8. Ans. (A)

Sol. (A) $\frac{1}{2}(P_0 + 2P_0)4V_0 = 6P_0V_0$

(B) AB $P \propto V$

$T \propto V^2$

$\frac{T_A}{T_B} = \left(\frac{4V_0}{8V_0}\right)^2$

$T_B = 400$ K.

$\Delta UAB = nCUDT$

$$-\frac{3}{25}\frac{3}{2}\left(\frac{25}{3}\right) \times 300 = 450J$$

(C) WCYCLIZ = _____

$$-(4P_0)7V_0 + \frac{1}{2}(R_0 + 4R_0)3V_0 + P_0V_0$$

$$-\left(28 + \frac{15}{2} + 1\right)P_0V_0 = \frac{-39}{2}P_0V_0$$

(D) UCD = $U_{CD} = nCv\Delta T = \frac{3}{25}\frac{3}{2}(-700)$

$T_B = 400$ L

$T_C = 800$ K

$T_B = 100$ K

CD = $V \propto T$

9. Ans. (A)

Sol. $F = \frac{mv^2}{r} = \frac{ke^2}{r^4}$ .... (1) $v^2 \propto \frac{1}{r^3}$

$mvr = \frac{nh}{2\pi}$ .... (2) $vr \propto n$

$v^2 \propto \frac{v^3}{n^3}$ $\boxed{v \propto n^3}$

$r \propto n^{-2}$

(A) $I = \frac{dq}{dt} = \frac{e}{T} = \frac{e}{2\pi r}v$

$I \propto \frac{v}{r} \propto \frac{n^3}{n^{-2}}$

(B) $B = \frac{\mu_0 I}{2\pi r} \propto \frac{n^5}{n^{-2}}$

(C) $L = mvr \propto n$

(D) $\mu = NIA \propto n^5 n^{-4}$

$\mu \propto n$

10. Ans. (B)

Sol. $U = U_0 \ln\left(\frac{r}{r_0}\right)$

$F = \frac{mv^2}{r} = \frac{U_0 r_0}{r}\frac{1}{r_0} = \frac{U_0}{r}$

$v \propto r^0$

$mvr = \frac{nh}{2\pi}$

$vr \propto n$

$r \propto n$

$v \propto n^0$

(A) $I \propto \frac{v}{r} = n^{-1}$

(B) $B \propto \frac{I}{r} \propto \frac{v}{r^2} \propto n^{-2}$

(C) $L = mvr \propto n$

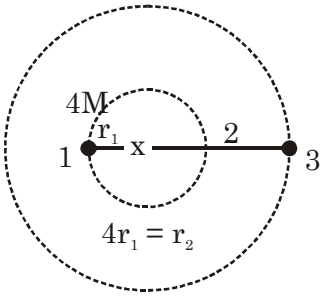
(D) $\mu = NIA \propto \frac{v}{r}r^2 \propto vr$

$\mu \propto n$

SECTION-II

1. Ans. 0.25

Sol.

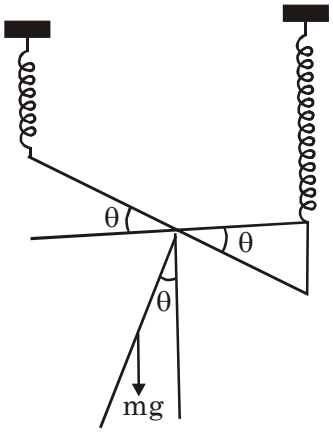


$\frac{K.E_{4m}}{K.E_m} = \frac{\frac{1}{2}4mv_1^2}{mv_2^2}$

$= \frac{4r_1^2w^2}{r_2^2w^2} = \frac{4r_1^2}{r^2} = \frac{1}{4}$

2. Ans. 12.00

Sol. Let the body be displaced through an angle θ about its mean position.



Net torque on the body at this position about C.

$\tau = -\left(2Ka\theta a\cos\theta + mga\sin\theta\right)$

$= -\left(2Ka^2 + mga\right)\theta$

(where θ is small, $\sin\theta = \theta$ and $\cos\theta = 1$)

Angular acceleration

$$\alpha = \frac{\tau}{I} = -\frac{-\left(2Ka^2 + mga\right)\theta}{\frac{m\times(2a)^2}{12} + \frac{m\times(2a)^2}{3}}$$

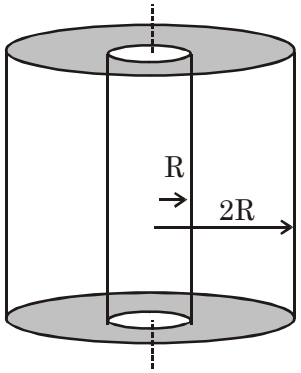
or $\alpha = -\frac{-\left(2Ka^2 + mga\right)}{\frac{5ma^2}{3}}$

$\Rightarrow \omega = \sqrt{\frac{\left(2Ka^2 + mga\right)}{5ma^2/3}} = \sqrt{\left(\frac{6K}{5m} + \frac{3g}{5a}\right)}$

$= \sqrt{\frac{6}{5}\times\frac{96}{6} + \frac{30}{5\times5}\times104} = 12 \text{ rad/s}$

3. Ans. 6.00

Sol.



$$H = -K,\ 2\pi rl\frac{dT}{dr}$$

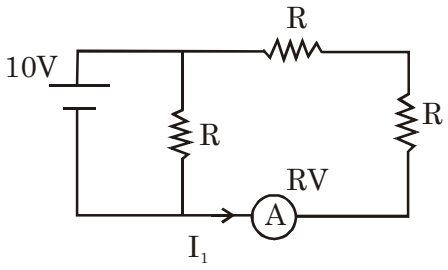
$$\int_{R_1}^{R_2}\frac{H\,dr}{2\pi rl} = -K\int_{T_1}^{T_2} dT$$

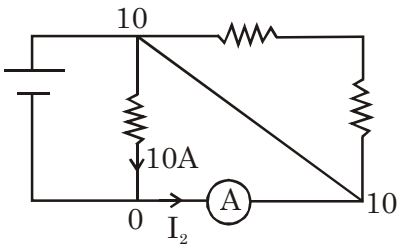
$$H = \frac{2\pi / k\left(T_1 - T_2\right)}{\ln\frac{R_2}{R_1}}H_i = H_f$$

$\therefore$ n = 6

4. Ans. 1.33

Sol.





$I_2 = 4I_1$

$$\frac{10}{RV} = \frac{4(10)}{(4 + RV)}$$

4 + RV = 4RV

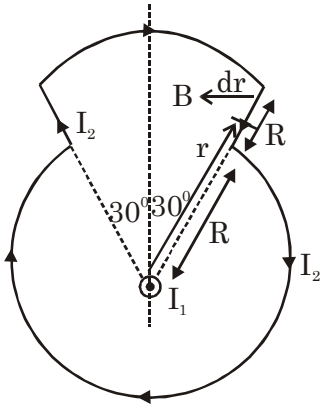
3RV = 4

RV = 1.33

5. Ans. 2.00

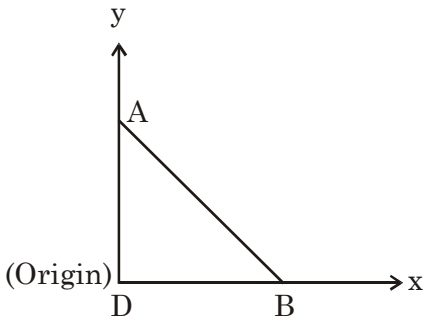
Sol. $$\int d\tau = 2\int_R^{2R}\frac{\mu_0 I_1 I_2}{2\pi x}\times\sin 30^0\,dx$$

τ = 20



6. Ans. 6.66 to 6.67

Sol.



Potential on vertex of cube of side a is V_0

Let $V \propto \frac{1}{4\pi\varepsilon_0}\rho^x a^y$

Dimensionally x = 1, y = 2

If side is made half potential at vertex $= \frac{V_0}{4}$

Net potential at centre due to eight cubes of side $\frac{a}{2} = 8\times\frac{V_0}{4}$

$$8\times\frac{V_0}{4} = 2V_0$$

7. Ans. 3.00

Sol. $\ell = 16.2 \pm 0.1$ cm

$b = 10.1 \pm 0.1$ cm

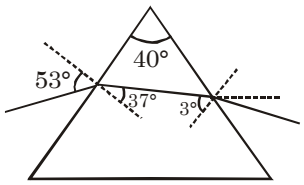
$\ell b = 163.62 \pm 2.6\ cm^2$

using proper rules

$\ell b = 164 \pm 3\ cm^2$

8. Ans. 17.00

Sol.



$$1\sin 53 = \frac{4}{3}\sin r_1$$

$$\frac{4}{5} = \frac{4}{3}\sin r_1$$

$\Rightarrow r_1 = 37° \Rightarrow r_2 = 3°$

$$\frac{4}{3}\times\sin 3° = \text{sine} \Rightarrow e = 4°$$

$\delta = i + e - A$

$= 53 + 4 - 40$

$= 17°$

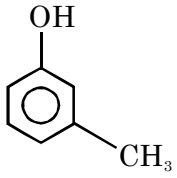
PART–2 : CHEMISTRY SOLUTION

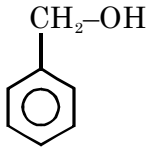
SECTION-I

1. Ans. (B,C)

2. Ans.(B,C)

3. Ans. (B)

Sol.  Functional group is phenol.

 Functional group is Alcohol.

4. Ans. (A,C)

Sol. $X^+ + H_2O \rightleftharpoons XOH + H^+$

$$K_H = \frac{k_w}{k_b} = \frac{10^{-14}}{4\times10^{-5}} = 2.5\times10^{-10}$$

$CH^2 \simeq k_H$

$(H^+) = Ch = \sqrt{c\cdot k_H}$

$10^{-5} = \sqrt{c\times2.5\times10^{-10}}$

$C = 0.4$ molar

$$0.4 = \frac{80}{M_{xy}}\times\frac{1}{2} \Rightarrow M_{xy} = \frac{40}{0.4} = 100$$

$$h = \sqrt{\frac{k_H}{C}} = \sqrt{\frac{2.5\times10^{-10}}{0.4}}$$

$h = 2.5\times10^{-5}$

$\%h = 2.5\times10^{-3}$

$$r_{x^+} + r_{y^-} = \sqrt{3}\frac{a}{2}$$

$$1.6 + 1.864 = r_{x^+} + r_{y^-} = \frac{\sqrt{3}}{2}a$$

$a = 4Å$

$$\text{density} = \frac{1\times\frac{100}{N_0}}{(4\times10^{-8})^3} \text{ gm/cc}$$

$= 2.6$ gm/c.c.

5. Ans. (A,B,D)

6. Ans. (A,B,C,D)

7. Ans. (B)

Sol. P–(3); Q–(1),(5); R–(1),(2),(5); S–(3)

8. Ans. (C)

9. Ans. (B)

10. Ans. (C)

Sol.

(P) $$E_{cell} = -\frac{0.0591}{2}\log\frac{C}{2C} = 8.89\times10^{-3}$$

Spontaneous, cell works, conc. cell.

(Q) $$E_{cell} = -0.0591\log\frac{1}{1} = 0$$

Non spontaneous, cell will not work conc. cell.

(R) $$E_{cell} = -\frac{0.0591}{2}\log\frac{0.01}{0.1^2} - 0.74 + 0.8 = 0.06$$

Spontaneous, cell works.

(S) $$E_{cell} = -\frac{0.0591}{1}\log\frac{Cl^-}{K_{sp}} - \frac{0.0591}{1}\log\frac{1}{[Ag^+]}$$

$$= \frac{0.0591}{2}\log\frac{0.1\times0.1}{K_{sp(AgCl)}} = +ve$$

Spontaneous, cell works, conc. cell.

at anode,

$Ag(s) + Cl^-(aq) \rightarrow AgCl(s) + e^-$

SECTION-II

1. Ans. (5.66)

Sol.: [Range 5.5 to 5.7]

$$PI = \frac{2.2+9.1}{2} = \frac{11.3}{2} = 5.66$$

2. Ans. (6.00)

Sol.: 2, 3, 4, 6, 7, 8

With coordination number six and oxidation state +2, wo (n – 1)d orbitals are not available for d^2sp^3.

3. Ans. (9.10 to 9.40)

Sol.: $z = 1 + \frac{Pb}{RT}$

$$\frac{dz}{dp} = \frac{1}{10} = \frac{L}{RT}$$

$$b = \frac{RT}{10} = \frac{22.4}{10} = 2.24$$

$$4 \times V \times N_0 = 2.24$$

$$V = \frac{2.24}{4 \times 6} \times 10^{-23}\ \ell$$

$$= \frac{2.24 \times 1000}{24} \times 10^{-23}\ cm^3$$

$$= 9.33 \times 10^{-22}\ cm^3$$

4. Ans. (36.00)

Sol.: (a) ZrI_4 Oxidation state = 4

(b) $[Ni(CO)_4]$ Bond order = 3

(c) $Na_2\ [Zn(CN)_4]$ = 3

5. Ans. (6.00)

Sol. All aldehyde hemeacetal α-hydroxy keto group, HCHO, can be oxidised by Tollen's regent.

(i), (ii), (iii), (v), (vi), (viii)

6. Ans. (0.18 to 0.20)

Sol.:

$$HNO_3 + 2H_2SO_4 \rightleftharpoons NO_2^+ + H_3O^+ + 2HSO_4^-$$

1	100			
0.7	99.7	0.3	0.3	0.6
	$\cong$ 100			

Total moles of particles = 0.7 + 0.3 + 0.3 + 0.6

= 1.9

$$\therefore \quad m = 1.9 \times \frac{1000}{9800} \simeq 0.19$$

7. Ans. (4.00)

Sol.:

(A) $Ag_2CrO_4\downarrow$ (yellow) + $NH_3 \rightarrow$

$$[Ag(NH_3)_2]^+ + CrO_4^{2-}$$

(B) $Cu(OH)_2 \cdot CuSO_4\downarrow$(blue) + $8NH_3 \rightarrow$

$$2[Cu(NH_3)_4]^{2+} + SO_4^{2-} + 2OH^-$$

(C) $PbSO_3\downarrow$ (white) + $4OH^- \rightarrow$

$$[Pb(OH)_4]^{2-} + SO_3^{2-}$$

Soluble in sodium hydroxide not in NH_3.

(D) Soluble in sodium hydroxide not in NH_3.

(E) $Ni(OH)_2\downarrow$ (green) + $6NH_2 \rightarrow [Ni(NH_3)_6]^{2+}$ (deep blue coloration) + $2OH^-$

(F) $Zn_2(PO_4)_2 \downarrow$ (white) + $12NH_3 \rightarrow 3[Zn(NH_3)_4]^{2+} + 2PO_4^{3-}$

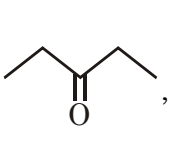
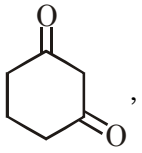
(G) Insoluble in NH_3, soluble appreciably in boiling concentrated H_2SO_4.

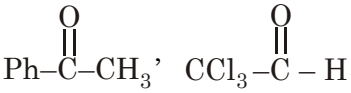
(H) $Bi(OH)_2NO_3$ is insoluble in NH_3.

(I) $Mn(OH)_2$ is insoluble in excess conc. solution of ammonia

8. Ans. (5.00)

Sol. Chloroform given by

$CH_3 - CH_2 - OH$,  ,  ,

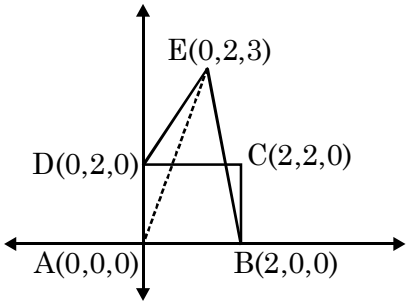


PART-3 : MATHEMATICS — SOLUTION

SECTION-I

1. Ans. (A,C)

Sol. Plane ABE is $\begin{vmatrix} x & y & z \\ 0 & 2 & 3 \\ 2 & 0 & 0 \end{vmatrix} = 0$



$x(0) - y(-6) + z(-4) = 0$

$\Rightarrow 3y - 2z = 0$

2. Ans. (A,C,D)

Sol. z_1, z_2 lie on the circle whose centre is (–2, –3) and radius is equal to.

$\because \arg(z_3 - z_1) = \arg(z_3 - z_2)$

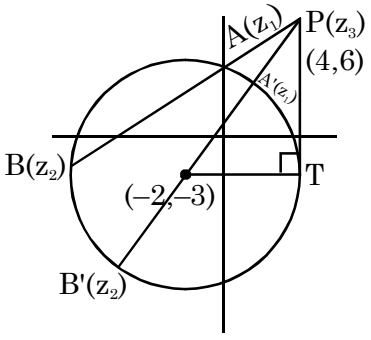
$\Rightarrow z_1$, z_2 and z_3 are collinear

$PA \cdot PB = PT^2$

$|z_3 - z_1|\,|z_3 - z_2| = OP^2 - OT^2$

$= 36 + 81 - 16$

$= 101$



$|z_1 - z_2|$ is maximum $\Rightarrow z_1$ and z_2 are extremities of diameter, $|z_1| = 4 - \sqrt{13}$ and $z_2 = 4 + \sqrt{13}$

$\therefore |z_1 z_2| = 16 - 13 = 3$

$|z_1| + |z_2| = 8$

3. Ans. (B,C)

Sol. $(A^{-1}BA)^{-1} = 2(\text{adj } A)$

$A^{-1}BA = 2 \text{ adj } A$

$BA = 2A(\text{adj } A) = 2|A|\, I_3$

$B = 2|A|A^{-1} = 10A^{-1}$

$\therefore \det(B) = 10^3 \times \frac{1}{5} = 200$

$\text{Tr.}(B) = 10\ \text{Tr.}(A^{-1}) = 10 \times 3 = 30$

4. Ans. (B,D)

Sol. As tangents are perpendicular to each other and its points of intersections $P\left(\frac{7}{5}, -\frac{4}{5}\right)$ lies on directrix. AB is focal chord.

Equation of AB is $x - y = 4$, as focus is (α, β) so $\alpha - \beta = 4 \Rightarrow \beta = \alpha - 4$

Now $\angle ASP = 90°$

$$\Rightarrow \frac{-2 - \alpha + 4}{2 - \alpha} \cdot \frac{-\frac{4}{5} - \alpha + 4}{\frac{7}{5} - \alpha}$$

$$\Rightarrow \frac{2 - \alpha}{2 - \alpha} \cdot \frac{\frac{16}{5} - \alpha}{\frac{7}{5} - \alpha} = -1$$

$$\Rightarrow \frac{16}{5} - \alpha = \alpha - \frac{7}{5};\ \alpha = \frac{23}{10}$$

$$\Rightarrow \text{Focus is } \left(\frac{23}{10}, -\frac{17}{10}\right)$$

Also, $\ell_1 = AS = \frac{3\sqrt{2}}{10}$, $\ell_2 = BS = \frac{27\sqrt{2}}{10}$

$$\frac{1}{a} = \frac{1}{\ell_1} + \frac{1}{\ell_2} \Rightarrow a = \frac{27\sqrt{2}}{100} \Rightarrow 4a = \frac{27\sqrt{2}}{25}$$

5. Ans. (A,C)

Sol. $I_0 = \int_0^{\pi/4} \underbrace{\left(x-\frac{\pi}{4}\right)^{50}}_{I} \underbrace{\sec^2 x\,dx}_{II}$

$$I_0 = \left[\left(x-\frac{\pi}{4}\right)^{50}\cdot \tan x\right]_0^{\pi/4}$$

$$-50\int_0^{\pi/4}\left(x-\frac{\pi}{4}\right)^{49}\cdot \tan x\,dx$$

$$\therefore \quad I_0 = -50\int_0^{\pi/4}\left(x-\frac{\pi}{4}\right)^{49}\tan x\,dx$$

$$\therefore \quad I_1 = \frac{\int_0^{\pi/4}\left(x-\frac{\pi}{4}\right)^{50}\sec^2 x\,dx}{\int_0^{\pi/4}\left(x-\frac{\pi}{4}\right)^{49}\cdot\tan x\,dx} = -50$$

$$I_2 = \frac{1}{100}\int_0^{\pi/4}\underbrace{\left(x-\frac{\pi}{4}\right)^{100}}_{I}\cdot\underbrace{\sec^2 x\,dx}_{II}$$

$$-\int_0^{\pi/4} x^{99}\left(\frac{1-\tan x}{1+\tan x}\right)dx$$

Apply King prop.,

$I_2 = 0$

$\therefore \quad I_1 < I_2$

$I_1 \cdot I_2 = 0$

6. Ans. (B,C)

Sol. $\frac{\cos A}{a} = \frac{b^2+c^2-a^2}{2abc}, \frac{\cos B}{b} = \frac{c^2+a^2-b^2}{2abc}, \frac{\cos C}{c} = \frac{a^2+b^2-c^2}{2abc}$

$$\therefore \frac{\cos A}{a}+\frac{\cos B}{b}+\frac{\cos C}{c} = \frac{a^2+b^2+c^2}{2abc}$$

$$\& \quad \frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3} = \frac{s-a}{\Delta}+\frac{s-b}{\Delta}+\frac{s-c}{\Delta}$$

$$= \frac{3s-2s}{\Delta} = \frac{s}{\Delta} = \frac{1}{r}$$

7. Ans. (A)

Sol.

$$\frac{d(g(x))}{d f(x)} = \begin{cases} \frac{-2}{1+(f(x))^2}, & f(x)\in(-1,1) \\ \frac{-2}{1+(f(x))^2}, & f(x)\in(-\infty,-1) \\ \frac{-2}{1+(f(x))^2}, & f(x)\in(1,\infty) \end{cases}$$

(P) $\frac{-2}{1+(f(x))^2} = -\frac{1}{13}$

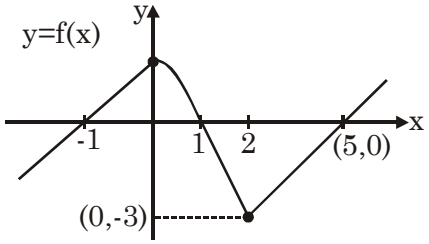
$\Rightarrow \quad 1 + (f(x))^2 = 26$

$\Rightarrow \quad f(x) = \pm 5$

$f(x) = 5$ when $x = 10$

$f(x) = -5$ when $x = -6$

(Q)



f(x) has local maxima at x = 0 and local minima at x = 2.

(R) From the graph of y = f(x) we can see that f(x) = –k has 2 positive and one negative root if $-k \in (-3, 1)$

$\Rightarrow k \in (-1, 3)$

(S) $g'(x) = \frac{-2}{1+(f(x))^2} \times f'(x)$

$g'(x) < 0 \Rightarrow f'(x) > 0$

$\Rightarrow x \in (-\infty, 0) \cup (2, \infty)$

8. Ans. (C)

Sol.

(P) $\frac{d^2y}{dx^2} = 6x - 4$

$\Rightarrow \frac{dy}{dx} = 3x^2 - 4x + c$

at $x = 1$, $\frac{dy}{dx} = 0 \Rightarrow c = 1$

$\therefore \frac{dy}{dx} = 3x^2 - 4x + 1$

$\Rightarrow y = x^3 - 2x^2 + x + d$

at $x = 1, y = 5$

$\Rightarrow 5 = 1 - 2 + 1 + d \Rightarrow d = 5$

$\therefore f(x) = x^3 - 2x^2 + x + 5$

$f(0) = 5$

(Q) $\frac{dy}{dx} = 2ax + b$

$\left.\frac{dy}{dx}\right|_{(1,2)} = 2a + b$

$y = ax^2 + bx + \frac{7}{2}$ passes through (1, 2)

$\Rightarrow 2 = a + b + \frac{7}{2}$

$\Rightarrow \boxed{a + b = \frac{-3}{2}}$ .....(1)

$y = x^2 + 6x + 10$

$\frac{dy}{dx} = 2x + 6$

Slope of normal at (–2, 2) $= \frac{-1}{2(-2)+6}$

$= -\frac{1}{2}$

$\therefore \boxed{2a + b = -\frac{1}{2}}$ .....(2)

From (1) and (2)

$a = 1, b = \frac{-5}{2}$

$\therefore \frac{a}{2} - b = \frac{1}{2} + \frac{5}{2} = 3$

(R) $5\alpha(3x^2) + 10\alpha(2x) + 1 + 2\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx} = \frac{-15\alpha^2x^2 - 20\alpha x - 1}{2}$

$\Rightarrow \frac{dy}{dx}$ at $A = \frac{-1}{2}$ A : (0, 2)

Equation of normal at A is

$y - 2 = 2(x - 0)$

$\Rightarrow y = 2x + 2$

Let normal meets the curve at B.

$\Rightarrow 5\alpha^2x^3 + 10\alpha x^2 + x + 4x + 4 - 4 = 0$

$\Rightarrow 5x(\alpha x + 1)^2 = 0$

$\Rightarrow x = \frac{-1}{\alpha}$

$\therefore B : \left(-\frac{1}{\alpha}, \frac{-2}{\alpha} + 2\right)$

Slope of tangent at B

$= \frac{-\frac{15\alpha^2}{\alpha^2} + \frac{20\alpha}{\alpha} - 1}{2} = 2$

$\therefore$ Equation of tangent is

$y + \frac{2}{\alpha} - 2 = 2\left(x + \frac{1}{\alpha}\right)$

$\Rightarrow \boxed{y = 2x + 2}$

(S) $f(x) = \frac{px}{e^x} - \frac{x^2}{2} + x$

$f'(x) = -px\,e^{-x} + pe^{-x} - x + 1$

$f'(x) = pe^{-x}(-x + 1) + (-x + 1)$

$= (pe^{-x} + 1)(1 - x) \le 0, \ \forall \ x < 0$

$\Rightarrow p \le -1$

9. Ans. (A)

Sol.

(P) $f(x) = \text{sgn}(\sin^2 x - \sin x - 1)$

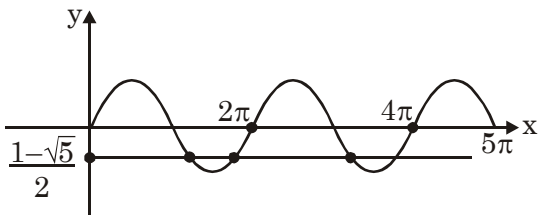
$\sin^2 x - \sin x - 1 = 0$

$\Rightarrow \sin x = \frac{1 \pm \sqrt{5}}{2}$

$\Rightarrow \sin x = \frac{1 - \sqrt{5}}{2}$

f(x) will be discontinuous when

$\sin x = \frac{1 - \sqrt{5}}{2}$



$\Rightarrow x \in (0, 4\pi]$ or $x \in (0, 5\pi]$

(Q) $y = \left[\frac{f(x)}{g(x)}\right]$

$y = \left[\frac{x^2 - x + k - 2}{x^2 - x + 1}\right]$

$y = \left[1 + \frac{k-3}{x^2 - x + 1}\right]$

$y = 1 + \left[\frac{k-3}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}}\right]$

for y to be continuous in R

$0 \le k - 3 < \frac{3}{4} \Rightarrow k \in \left[3, \frac{15}{4}\right)$

$\Rightarrow a = 3,\ b = \frac{15}{4}$

$b - \frac{5a}{4} = \frac{15}{4} - \frac{15}{4} = 0$

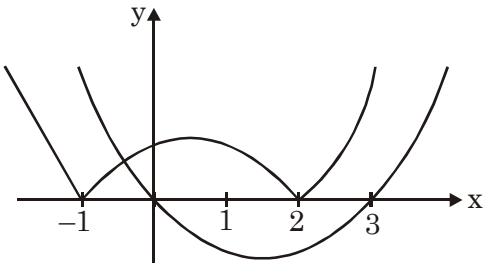
(R) $f(x) = \begin{cases} \cos^{-1} x, & \frac{1}{2} \le |x| \le 1 \\ \frac{2|x|}{3} + \frac{\pi}{3} - \frac{1}{3}, & |x| < \frac{1}{2} \end{cases}$

$f(x) = \begin{cases} \cos^{-1} x, & x \in \left[-1, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 1\right] \\ -\frac{2x}{3} + \frac{\pi}{3} - \frac{1}{3}, & x \in \left(-\frac{1}{2}, 0\right) \\ \frac{2x}{3} + \frac{\pi}{3} - \frac{1}{3}, & x \in \left[0, \frac{1}{2}\right) \end{cases}$

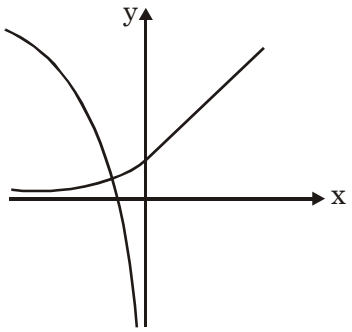
$\cos^{-1} x$ is non differentiable at x = –1, 1.

f(x) is non-differentiable at x = –1, $-\frac{1}{2}$, 0, $\frac{1}{2}$, 1.

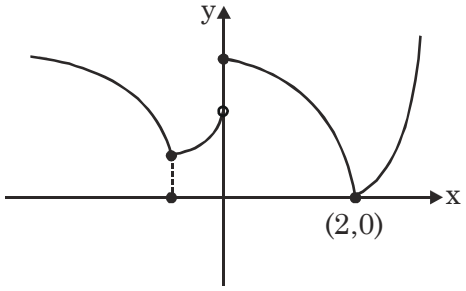
(S) max $[|x^2 - x - 2|, x^2 - 3x]$



max $(\ln(-x), e^x)$



y = f(x)



From the above graph of y = f(x) we can see that f(x) is non differentiable at 3 points.

10. Ans. (A)

Sol.

(P) If a = 1

$f(x) = \lim_{t \to \infty} \frac{(1 + \sin \pi x)^t - 1}{(1 + \sin \pi x)^t + 1}$

If $x \in (0, 1) \cup (2, 3) \cup (4, 5)$

f(x) = 1

If $x \in (1, 2) \cup (3, 4) \cup (5, 6)$

f(x) = –1

If x = 2, 3, 4, 5

f(x) = 0

$\therefore$ f(x) is discontinuous at x = 1, 2, 3, 4, 5

(Q) If a = 3

$f(x) = \lim_{t \to \infty} \frac{(3 + \sin \pi x)^t - 1}{(3 + \sin \pi x)^t + 1}$

$f(x) = 1\ \forall\ x \in R$

$\therefore$ f(x) is continuous $\forall\ x \in R$

(R) If a = 0.5

$f(x) = \lim_{x \to \infty} \frac{(0.5 + \sin \pi x)^t - 1}{(0.5 + \sin \pi x)^t + 1}$

Limit does not exist at

$x = \frac{1}{6}, \frac{5}{6}, \frac{13}{6}, \frac{17}{6}, \frac{25}{6}, \frac{29}{6}$

(S) If a = 0

$$f(x)=\lim_{t\to\infty}\frac{(\sin\pi x)^t-1}{(\sin\pi x)^t+1}$$

(i) If $x=\frac{1}{2},\frac{5}{2},\frac{9}{2}$

f(x) = 0

(ii) If $x\in\left(0,\frac{1}{2}\right)\cup\left(\frac{1}{2},\frac{3}{2}\right)\cup\left(\frac{3}{2},\frac{5}{2}\right)\cup\left(\frac{5}{2},\frac{7}{2}\right)$

$$\cup\left(\frac{7}{2},\frac{9}{2}\right)\cup\left(\frac{9}{2},\frac{11}{2}\right)\cup\left(\frac{11}{2},6\right)$$

f(x) = –1

(iii) If $x=\frac{3}{2},\ \frac{7}{2},\ \frac{11}{2}$

f(x) is not defined

$\therefore$ f(x) is discontinuous at

$x=\frac{1}{2},\frac{3}{2},\frac{5}{2},\frac{7}{2},\frac{9}{2},\frac{11}{2}$

SECTION-II

1. Ans. (1.00)

Sol.: $(\vec{r}-\vec{a})\times(\vec{b}-\vec{a})$ is parallel to $\hat{n}$

$\Rightarrow (\vec{r}-\vec{a})\times(\vec{b}-\vec{a})=t\hat{n}$ ...(i)

$\therefore \left|(\vec{r}-\vec{a})\times(\vec{b}-\vec{a})\cdot\hat{n}\right|=t$

$$\left|\left[\vec{r}-\vec{a}\ \ \vec{b}-\vec{a}\ \ \hat{n}\right]\right|=|t| \quad ...(2)$$

From (1)

$$\vec{r}\times(\vec{b}-\vec{a})-\vec{a}\times\vec{b}=t\hat{n}$$

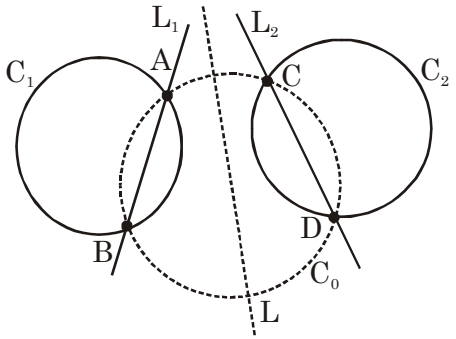
or $\left|\vec{r}\times(\vec{b}-\vec{a})-(\vec{a}\times\vec{b})\right|=|t|$ ...(3)

From (2) and (3)

$$\left|\left[\vec{r}-\vec{a}\ \ \vec{b}-\vec{a}\ \ \hat{n}\right]\right|=\left|\vec{r}\times(\vec{b}-\vec{a})-(\vec{a}\times\vec{b})\right|$$

2. Ans. (4.00)

Sol.: We represent the given situation through the following diagram :



- C_0 is the circle passing through A, B, C and D.
- L is the radical axis of C_1 and C_2 given by.

$S_1 - S_2 = 0$

$\Rightarrow 2x - 2y - 6 = 0$

We use the fact that the radical axes of three circles taken two at a time are concurrent. Thus, L, L_1 and L_2 are concurrent :

$$\begin{vmatrix}2 & -2 & -6\\ 1 & 2 & 3\\ 2 & 3 & \lambda\end{vmatrix}=0 \ \Rightarrow\ \lambda=4$$

3. Ans. (1.00)

Sol. $\sin y\ \frac{dy}{dx} - 2\cos y\cos x = -\sin^2 x\cos x$

Put $-\cos y = z$

$\Rightarrow \sin y\frac{dy}{dx}=\frac{dz}{dx}$

$\frac{dz}{dx}$ + (2 cos x)z = $-\sin^2 x\cos x$

I.F. = $e^{\int 2\cos dx}=e^{2\sin x}$

$z\cdot e^{2\sin x}=-\int e^{2\sin x}\cos x\sin^2 x\,dx$

$\therefore\ z=-\cos y=\frac{-\sin^2 x}{2}-\frac{1}{4}+\frac{\sin x}{2}+ce^{-2\sin x}$

y = 0 at x = 0

$\Rightarrow -1=0-\frac{1}{4}+0+c$

$\Rightarrow c=-\frac{3}{4}$

$\therefore \cos y = \frac{1}{2}\sin^2 x - \frac{1}{2}\sin x + \frac{1}{4} + \frac{3}{4}e^{-2\sin x}$

$\therefore A + B + C + D = 1$

4. Ans. (6.00)

Sol.: Let $P(E_k) = \lambda K(K+1)$

$$\sum_{k=1}^{n} P(E_k) = 1$$

$$\Rightarrow \lambda \sum_{k=1}^{n} k(k+1) = 1$$

$$\Rightarrow \lambda = \frac{3}{n(n+1)(n+2)}$$

Let P(E) denote the probability that a coin selected at random is biased.

$$P(E_1/E) = \frac{P(E_1) \times P(E/E_1)}{\sum_{k=1}^{n} P(E_k) \times P(E/E_k)}$$

$$P(E_1/E) = \frac{2\lambda \times \frac{1}{n}}{\sum_{k=1}^{n} \lambda \cdot k \cdot (k+1) \cdot \frac{k}{n}}$$

$$= \frac{2}{\sum_{k=1}^{n} k^2(k+1)}$$

$$= \frac{24}{n(n+1)(n+2)(3n+1)}$$

$$\therefore F(n) = \frac{24}{n(n+1)(n+2)(3n+1)}$$

$$\sum_{r=1}^{k} F(r) \cdot (3r+1)$$

$$= \sum_{r=1}^{k} \frac{24}{r(r+1)(r+2)}$$

$$= 12 \sum_{r=1}^{k} \left(\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} \right)$$

$$= 12 \left(\frac{1}{1.2} - \frac{1}{(k+1)(k+2)} \right)$$

$$\therefore \lim_{k \to \infty} \sum_{r=1}^{k} F(r)(3r+1) = \frac{12}{2} = 6$$

5. Ans. (8.00)

Sol.: $\frac{1}{a}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots\ldots, \frac{1}{a_9}$ are in A.P.

$$d = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$$

Series is $\frac{1}{20}, \frac{2}{20}, \frac{3}{20}, \ldots, \frac{9}{20}$

$$D = \begin{vmatrix} 20 & \frac{20}{2} & \frac{20}{3} \\ \frac{20}{4} & \frac{20}{5} & \frac{20}{6} \\ \frac{20}{7} & \frac{20}{8} & \frac{20}{9} \end{vmatrix}$$

$D = (20)^3 \ |A|$

$$\Rightarrow \frac{D}{|A|} = 8000$$

6. Ans. (10.00)

Sol.: $|A|^{16} = 4^8.5^{16}$

$|A| = 10$

$x + y + z = 10$

7. Ans. (3.00)

Sol.: $\frac{1}{a_1 a_n} + \frac{1}{a_2 a_{n-1}} + \ldots\ldots + \frac{1}{a_n a_1}$

$$= \frac{2}{a_1 + a_n} \left[\frac{1}{a_1} + \frac{1}{a_2} + \ldots\ldots + \frac{1}{a_n} \right]$$

$$\Rightarrow 1 - \tan\theta = \frac{1}{1 + \sin 2\theta}(1 + \tan\theta)$$

$$\Rightarrow (\cos\theta - \sin\theta) = \frac{\cos\theta + \sin\theta}{(\cos\theta + \sin\theta)^2}$$

$\Rightarrow \cos 2\theta = 1 \Rightarrow \theta = k\pi$

8. Ans. (10.00)

Sol. $S_{100} = \frac{0}{\left({}^{100}C_0\right)^5} + \frac{1}{\left({}^{100}C_1\right)^5} + \ldots$ (i)

Also $S_{100} = \frac{100}{\left({}^{100}C_0\right)^5} + \frac{99}{\left({}^{100}C_1\right)^5} + \ldots.$ (ii)

$\therefore$ On adding (1) and (2), we get

$2S_{100} = 100t_{100}$

$$\Rightarrow \frac{t_{100}}{S_{100}} = \frac{2}{100}$$